

Poula Keroslos Labe

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PROFESSIONAL SUMMARY

Electronics and Electrical Communications Engineering student at Cairo University (GPA: 3.82) with hands-on experience in digital design, analog circuits, and embedded systems. Active researcher on an autonomous collision avoidance system. Committed to building practical skills aligned with embedded systems and analog/mixed-signal engineering.

EDUCATION

Cairo University – Faculty of Engineering 2022 – 2026 (Expected)

B.Sc. Electronics and Electrical Communications Engineering | Cumulative GPA: 3.82 | Last Semester GPA: 3.67

Relevant Coursework:

- Circuits I & II, Electronics, Logic Design, Data Structures, ODE, Probability & Statistics, Optics

WORKSHOPS & TRAINING

Digital Design Workshop – IEEE Cairo University Student Branch 2025

Verilog HDL | Multi-session program | Certificate in progress

- Gained foundational and applied knowledge of Verilog HDL for digital circuit design
- Used ModelSim/Questa for simulation and testbench verification

Implementations:

- 4-bit ALU – arithmetic and logic operations with directed and random testbench verification
- 64-bit arithmetic shift register – synchronous load, four shift modes (left/right, 1-bit and 8-bit)
- Mealy and Moore FSM sequence detectors – overlapping and non-overlapping variants

Analog Design Workshop – IEEE Helwan University Student Branch 2025

Cadence Virtuoso | Certificate Awarded

- Completed structured training in analog circuit design and simulation
- Gained working familiarity with Cadence Virtuoso for schematic entry and simulation
- Covered fundamental analog design concepts including amplifier topologies and biasing

PROJECTS

Digital Multimeter – ATmega328 Microcontroller 2024

- Built a three-mode instrument (voltmeter, ammeter, ohmmeter) with ADC calibration and LCD output
- Implemented mode-switching logic and circuit interfacing in embedded C

Ultrasonic Navigation Aid for the Visually Impaired – Arduino Nano 2024

- Designed an assistive device using ultrasonic sensing and vibration feedback for obstacle detection
- Implemented real-time distance measurement with adaptive haptic response

RESEARCH & ACADEMIC ACTIVITIES

Autonomous Vehicle Collision Avoidance – Research Day Presenter 2025

Limit Breakers Team | Supervised by Dr. Samah El-Tantawy, Cairo University

- Co-developed a LiDAR-based collision avoidance system using Kalman filtering and LQR optimal control
- Implemented dynamic Time-to-Collision (TTC) thresholds for adaptive safety response
- Designed the research poster presented at Cairo University Research Day
- Delivered a research presentation covering system architecture, methodology, and results

TECHNICAL SKILLS

Hardware & EDA Tools: Arduino IDE, Cadence Virtuoso (familiar), ModelSim/Questa, Multisim, KiCad, LTspice

Programming & HDL: Verilog HDL, C/C++, Python

Domains: Embedded Systems, Digital Design, Analog Circuits, Autonomous Systems, Microcontroller Programming

LANGUAGES

Arabic (Native) | English B2 – Ministry of Defense Language Institute | German A1 – Ministry of Defense Language Institute